

Beyond the Syntax: Mastering Agentic Coding

The Future of Software Engineering

Awwal Badru

PhD Candidate & MS, CS (United States) | M.Sc CS (Australia)

| B.Ed CS/Maths (Nigeria)

Research Areas:

Empirical Software Engineering
Software Security
Software Engineering Education
Computer Science Education



vs



vs



Workshop Outline

DAY 1: Foundations & Demo

- The Evolution of AI & Coding
- Coding Agents & Agentic Loop
- Demo: Context, Action & Verification
- IDE Setup and Resources
- Interactive Q & A

Workshop Outline

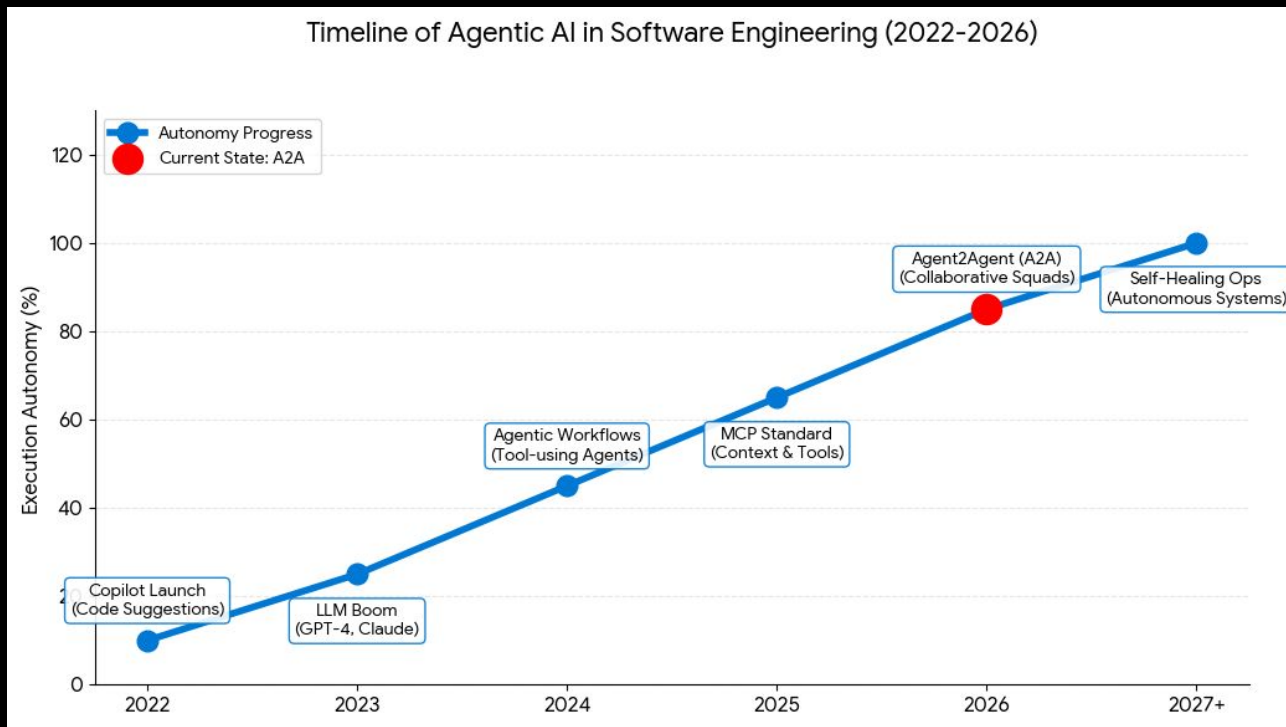
DAY 1: Foundations & Demo

- The Evolution of AI & Coding
- Coding Agents & Agentic Loop
- Demo: Context, Action & Verification
- IDE Setup and Resources
- Interactive Q & A

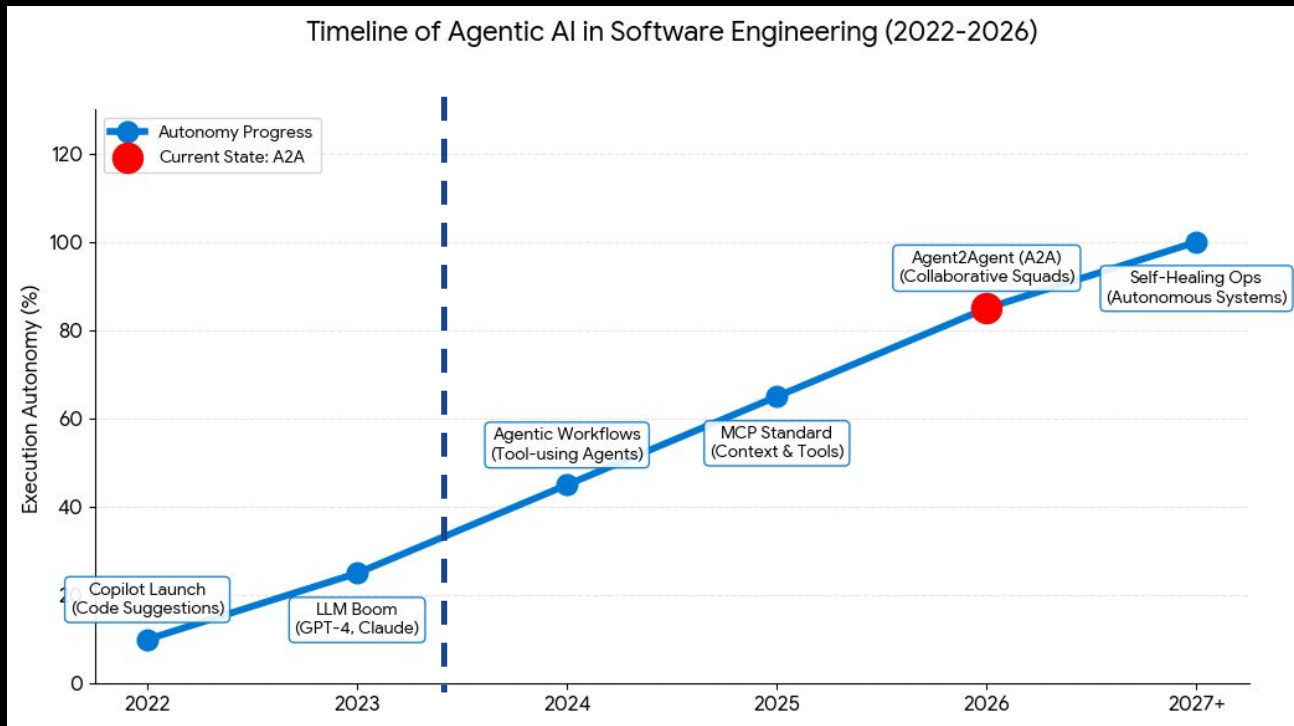
DAY 2: Practical Application

- Software Development Life Cycle
- Hands-on Building with Agents
- Git Versioning & Collaboration

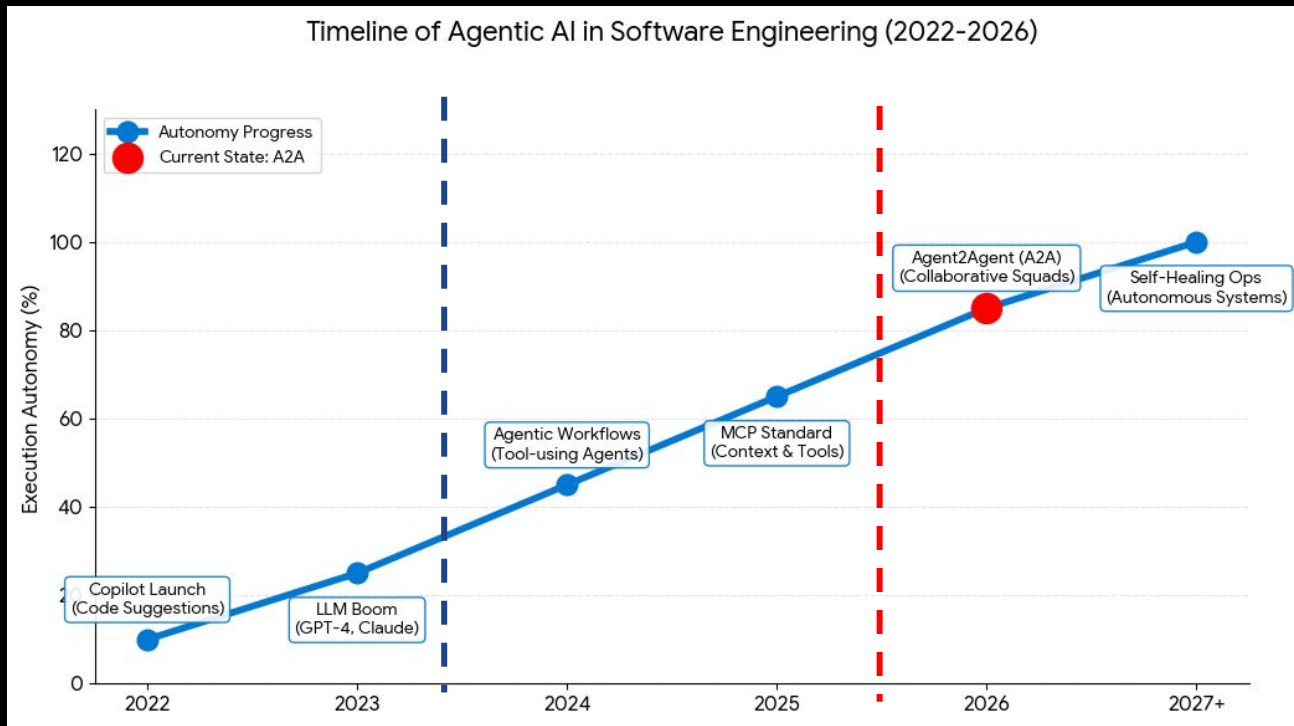
The Evolution of AI & Coding



The Evolution of AI & Coding



The Evolution of AI & Coding



The Evolution of AI & Coding

1

Code Completion

Intelligent snippets and autocomplete based on context. Tools like traditional IDE intellisense.



The screenshot shows a code editor with a file named 'hello_world.py'. The code is as follows:

```
1 def hello_world():  
2     print("Hello, World!")  
3  
4 if   
5     hello_world()
```

A red box highlights the 'hello_world()' suggestion in the autocomplete dropdown menu. The dropdown menu also shows 'Accept Word' and 'Accept' buttons.

The Evolution of AI & Coding

2

Code Generation

Generative AI creating entire functions and boilerplate from natural language prompts. The era of Copilots.

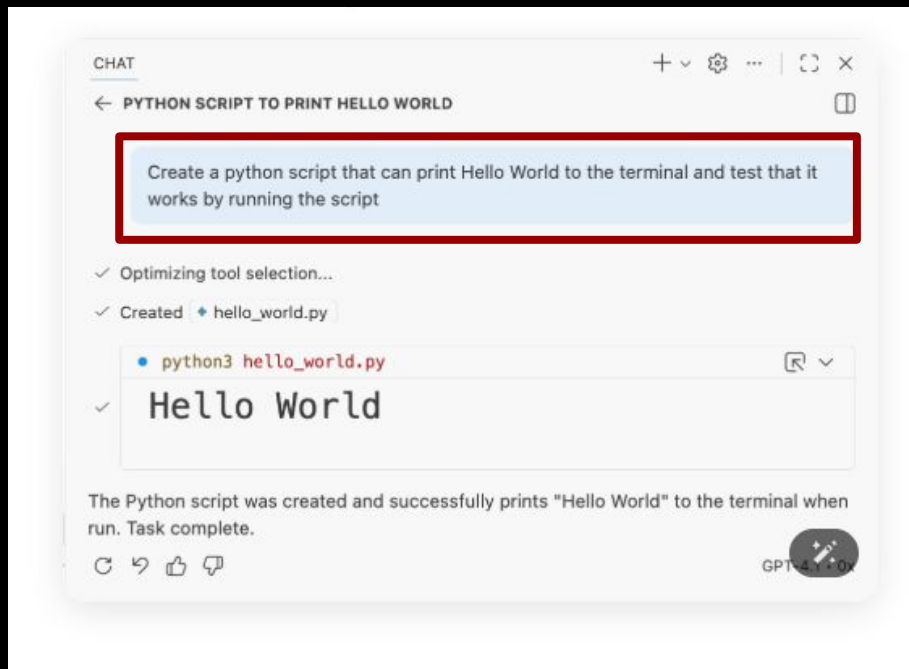


The Evolution of AI & Coding

3

Agentic Coding

Autonomous agents that reason, plan, and execute across files, using tools and verifying results.



Coding Agents

IDE

- Copilot in VS Code
- Cursor
- Windsurf
- Antigravity

Coding Agents

IDE

- Copilot in VS Code
- Cursor
- Windsurf
- Antigravity

CLI

- Copilot CLI
- Claude Code
- Pi

Coding Agents

IDE

- Copilot in VS Code
- Cursor
- Windsurf
- Antigravity

CLI

- Copilot CLI
- Claude Code
- Pi

Web

- Copilot in GitHub.com
- Codex
- Jules

Coding Agents



IDE

- Copilot in VS Code
- Cursor
- Windsurf
- Antigravity



CLI

- Copilot CLI
- Claude Code
- Pi



Web

- Copilot in GitHub.com
- Codex
- Jules



Custom

- Copilot SDK
- PyDantic AI
- LangChain

Coding Agents & Agentic Loop

What is an AI Agent?

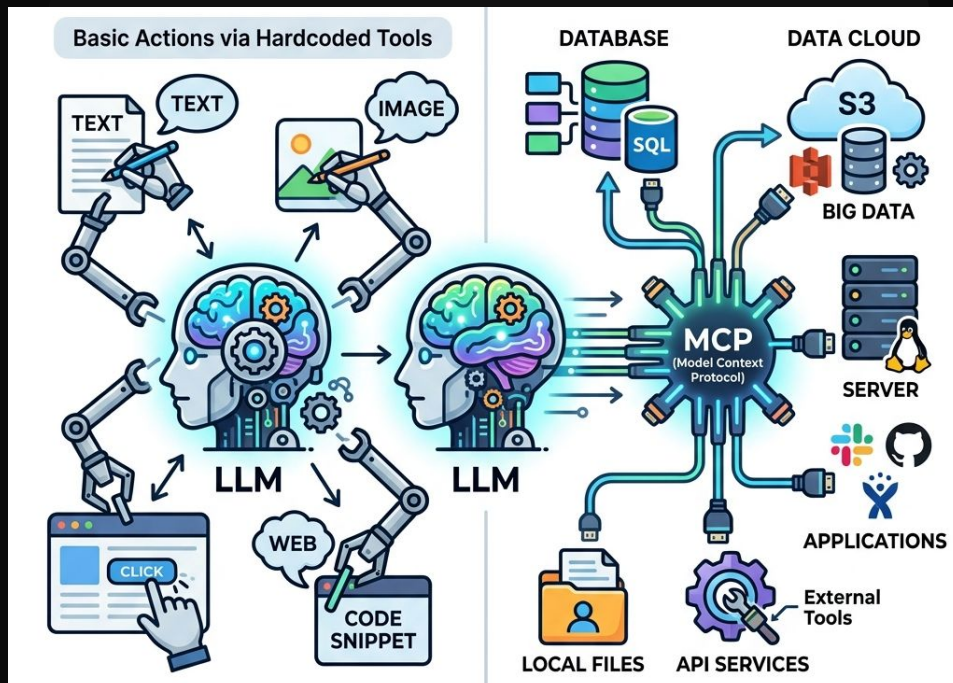
An AI Agent is an LLM that calls tools in a loop to achieve a goal.

Source: https://simonwillison.net/2025/Sep/18/agents/?utm_source=chatgpt.com

Coding Agents & Agentic Loop

What is an AI Agent?

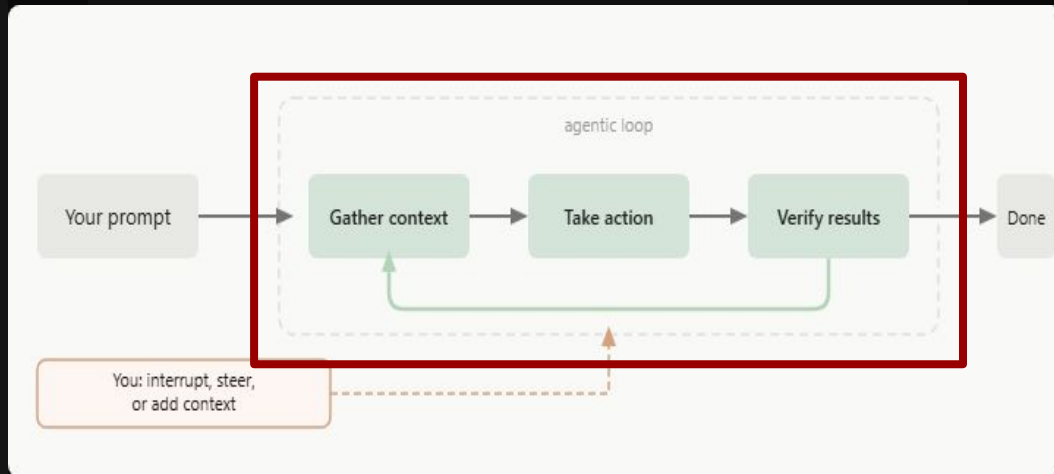
An AI Agent is an **LLM** that **calls tools** in a loop to achieve a goal.



Coding Agents & Agentic Loop

What is an AI Agent?

An AI Agent is an **LLM** that **calls tools** in a **loop** to achieve a goal.



Source: <https://code.claude.com/docs/en/how-claude-code-works>

Quick Pricing Overview (April 2026)

Service	Individual Plan	High-Usage/Max	Team/Business
GitHub Copilot	\$10 /mo	\$39 /mo (Pro+)	\$19 /user/mo
Claude	\$20 /mo	\$100 - \$200 /mo	\$25 /user/mo
Antigravity	\$29 /mo	\$99 - \$250 /mo	Custom
OpenAI Codex	\$20 /mo (via ChatGPT Plus)	\$200 /mo (Pro)	\$25 /user/mo

Demonstration

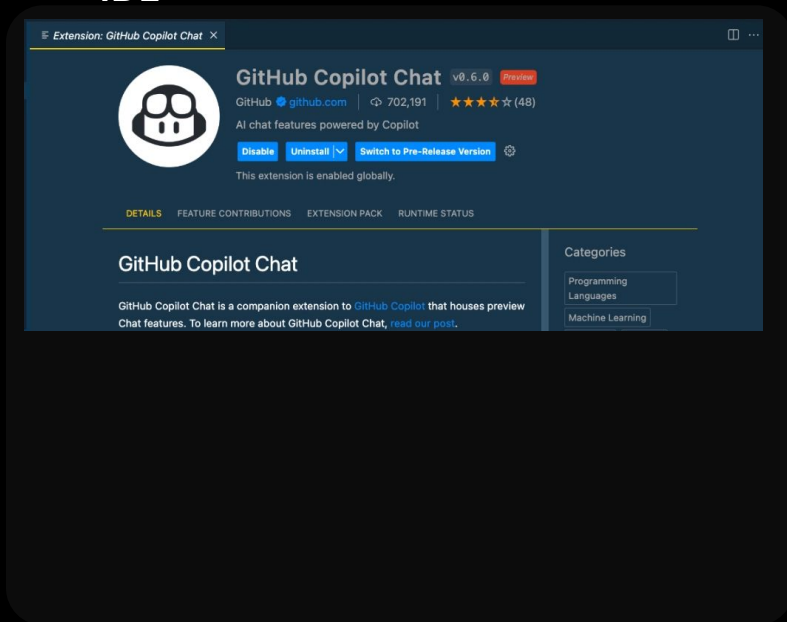
 IDE

 CLI

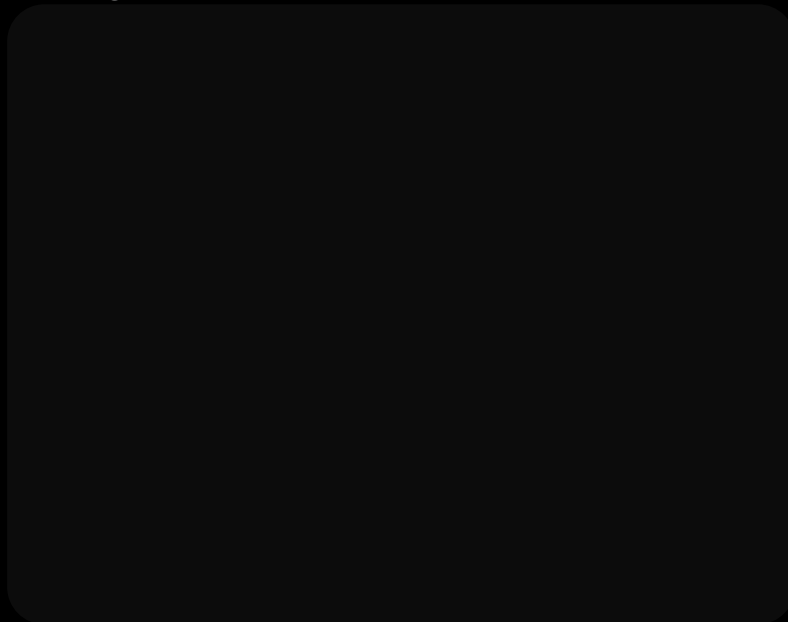
Demonstration



IDE



CLI



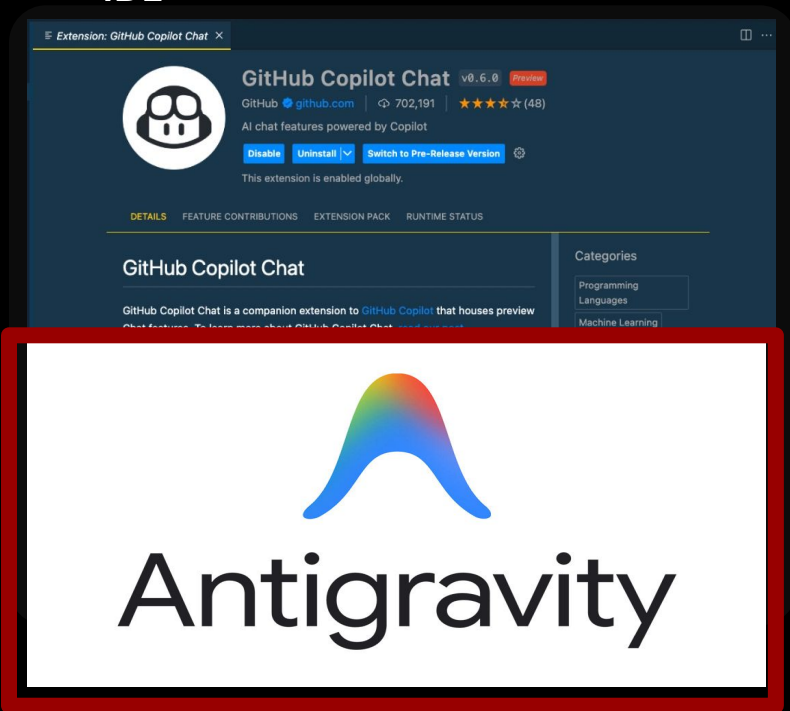
Demonstration



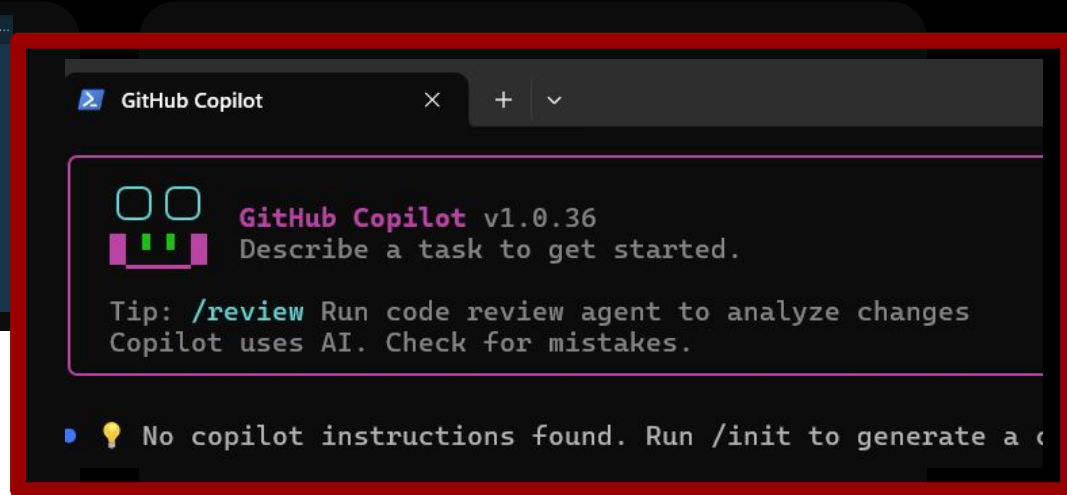
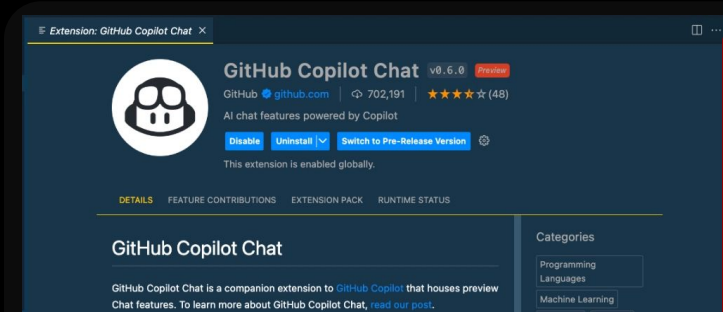
IDE



CLI



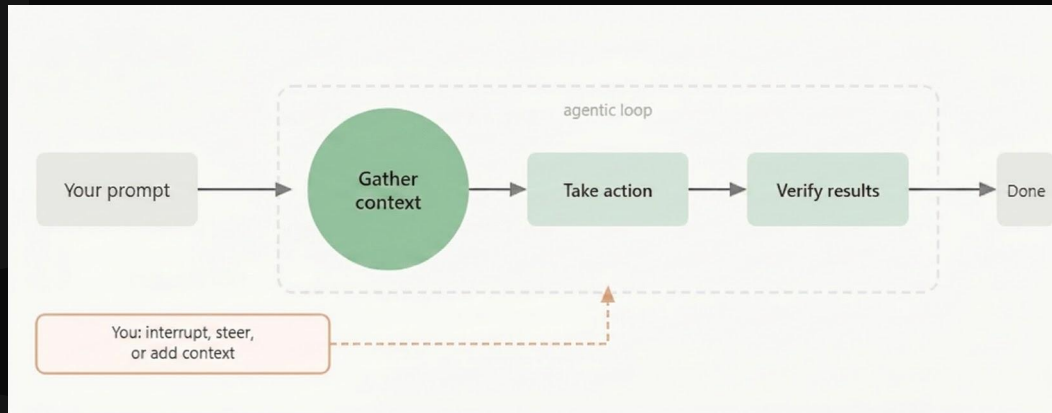
Demonstration



Equipping Agents with Context

Including Files and Assets

- Directly provide files, images, and GitHub items (Issues, PRs)
- Offer exemplar files containing best-practice code



Directive Documentation

- `copilot-instructions.md` — GitHub Copilot
- `cursor rules` — Specific to Cursor IDE
- `claude.md` — Tailored for Claude Code
- `agents.md` — The standard OpenAI format

Providing Tools to an Agent

MCP — Model Context Protocol

Open protocol for giving agents access to tools

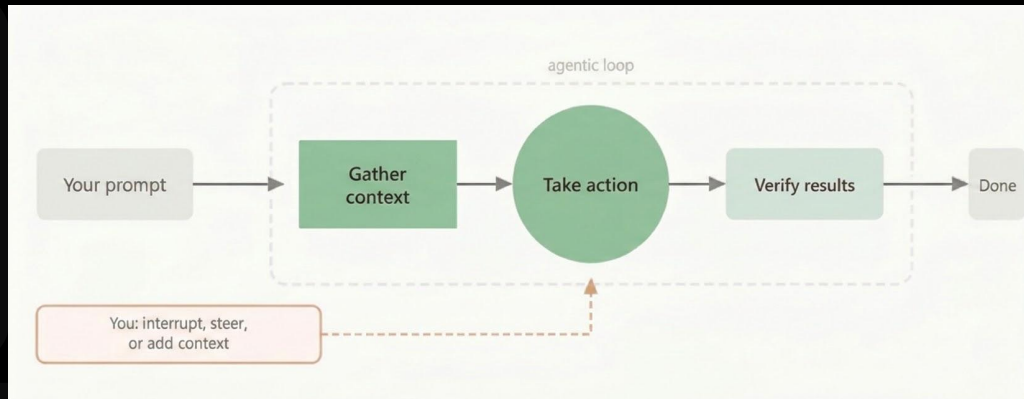
In IDE: search @mcp in extensions for vetted servers

Pro Tip: You can also create your own MCP servers

Agent Skills

A reusable workflow defined in a `SKILL.MD` file

Skills can include tools, MCP servers, scripts, and more!



Resources

IDE (VS Code)

- VS Code Installation
<https://code.visualstudio.com/download>
- Copilot Installation from VS Code extension
<https://www.youtube.com/watch?v=ws6t0NBle9g>
- Launching Copilot for free (please you need to have a github account)
https://www.youtube.com/watch?v=X_Aet9ndh_Y

IDE (Antigravity)

- Google Agent IDE Installation
<https://antigravity.google/download>

You need to have gmail account to sign in and start using the agent for free

Resources

CLI (Copilot)

- Installation npm (requires Node.js 22+)

<https://www.youtube.com/watch?v=BDxRhhs36ns>

- Installation on windows via (WinGet)

https://www.youtube.com/watch?v=3mCaZ_tYdI0

CLI (Copilot)

Install commands:

Windows (WinGet): `winget install GitHub.Copilot`

npm (requires Node.js 22+): `npm install -g @github/copilot`

macOS / Linux: `brew install copilot-cli`

Beyond the Syntax: Mastering Agentic Coding

The Future of Software Engineering

Awwal Badru

PhD Candidate & MS, CS (United States) | M.Sc CS (Australia)

| B.Ed CS/Maths (Nigeria)

Research Areas:

Empirical Software Engineering
Software Security
Software Engineering Education
Computer Science Education



vs



vs



Workshop Outline

DAY 1: Foundations & Demo

- The Evolution of AI & Coding
- Coding Agents & Agentic Loop
- Demo: Context, Action & Verification
- IDE Setup and Resources
- Interactive Q & A

DAY 2: Practical Application

- Software Development Life Cycle
- Hands-on Building with Agents
- Git Versioning & Collaboration

SDLC

The 7 Phases of SDLC

1. Planning

Define scope, costs, and identify risks.

SDLC

The 7 Phases of SDLC

1. Planning

Define scope, costs, and identify risks.

2. Requirements

Gather needs from stakeholders (the "what").

SDLC

The 7 Phases of SDLC

1. Planning

Define scope, costs, and identify risks.

2. Requirements

Gather needs from stakeholders (the "what").

3. Design

Architecture, UI/UX, and data models.

SDLC

The 7 Phases of SDLC

1. Planning

Define scope, costs, and identify risks.

2. Requirements

Gather needs from stakeholders (the "what").

3. Design

Architecture, UI/UX, and data models.

4. Implementation

The actual writing of the source code.

SDLC

The 7 Phases of SDLC

1. Planning

Define scope, costs, and identify risks.

2. Requirements

Gather needs from stakeholders (the "what").

3. Design

Architecture, UI/UX, and data models.

4. Implementation

The actual writing of the source code.



5. Testing

Verify software for bugs and requirements.

SDLC

The 7 Phases of SDLC

1. Planning

Define scope, costs, and identify risks.

2. Requirements

Gather needs from stakeholders (the "what").

3. Design

Architecture, UI/UX, and data models.

4. Implementation

The actual writing of the source code.



5. Testing

Verify software for bugs and requirements.

6. Deployment

Release to production environment.

SDLC

The 7 Phases of SDLC

1. Planning

Define scope, costs, and identify risks.

2. Requirements

Gather needs from stakeholders (the "what").

3. Design

Architecture, UI/UX, and data models.

4. Implementation

The actual writing of the source code.



5. Testing

Verify software for bugs and requirements.

6. Deployment

Release to production environment.

7. Maintenance

Fix issues, update, and monitor.

Hands-on Learning

- For UX/UI

<https://stitch.withgoogle.com/>

Hands-on Learning

What are app are you building?

1. Planning

Define scope, costs, and identify risks.

2. Requirements

Gather needs from stakeholders (the "what").

3. Design

Architecture, UI/UX, and data models.

4. Implementation

The actual writing of the source code.

5. Testing

Verify software for bugs and requirements.